

**Chapter 3. System of Differential Equations**  
**Sections 3.1, 3.2, 3.3, 3.4, 3.8, 3.9, 3.11, 3.12**

1. System of differential equations in  $n$  variables  $x_1, x_2, \dots, x_n$  is of the form

$$\begin{aligned}\frac{dx_1}{dt} &= f_1(t, x_1, \dots, x_n), \\ \frac{dx_2}{dt} &= f_2(t, x_1, \dots, x_n), \\ &\vdots \\ \frac{dx_n}{dt} &= f_n(t, x_1, \dots, x_n).\end{aligned}\tag{1}$$

We will often impose the initial conditions like

$$x_1(t_0) = x_1^0, x_2(t_0) = x_2^0, \dots, x_n(t_0) = x_n^0.$$

2. The  $n$ th-order linear differential equation is of the form

$$a_n(t) \frac{d^n y}{dt^n} + a_{n-1}(t) \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_1(t) \frac{dy}{dt} + a_0(t)y = g(t).\tag{2}$$

Let  $x_1(t) = y$ ,  $x_2(t) = \frac{dy}{dt}$ ,  $\dots$ ,  $x_n(t) = \frac{d^{n-1}y}{dt^{n-1}}$ , then the equation (2) can be transformed into the form of system of equations (1) as:

$$\begin{aligned}\frac{dx_1}{dt} &= x_2, \\ \frac{dx_2}{dt} &= x_3, \\ &\vdots \\ \frac{dx_{n-1}}{dt} &= x_n, \\ \frac{dx_n}{dt} &= \frac{g(t) - a_{n-1}(t)x_n - a_{n-2}(t)x_{n-1} - \dots - a_0(t)x_1}{a_n(t)}.\end{aligned}\tag{3}$$

3. Definition of a vector space (Page 273).
4. The vectors  $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^n$  spans  $\mathbf{V}$  if every element of  $\mathbf{V}$  can be expressed as a linear combination of  $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^n$ .
5. The vectors  $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^n$  are called linearly independent if  $c_1\mathbf{x}^1 + c_2\mathbf{x}^2 + \dots + c_n\mathbf{x}^n = 0$  implies  $c_1 = c_2 = \dots = c_n = 0$ ; otherwise, they are called linearly dependent.

6. System of homogeneous linear differential equations with constant coefficients:

$$\dot{\mathbf{x}} = \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad (4)$$

The set of all solutions of (4) forms a vector space  $\mathbf{V}$  with  $\dim(\mathbf{V}) = n$ .

- Let  $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^k$  be  $k$  solutions of (4). Then  $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^k$  are linearly independent if and only if  $\mathbf{x}^1(t_0), \mathbf{x}^2(t_0), \dots, \mathbf{x}^k(t_0)$  are linearly independent vectors in  $\mathbb{R}^n$  for a convenient  $t_0$ .
7. An nonzero vector  $\mathbf{v}$  satisfying  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$  is called a eigenvector of  $\mathbf{A}$  associated with eigenvalue  $\lambda$ .
- Characteristic polynomial of  $\mathbf{A}$  is given by

$$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}).$$

- To find all eigenvalues and corresponding eigenvectors  $(\lambda, \mathbf{v})$  of  $\mathbf{A}$ , one first finds all roots of  $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$ , and then for each eigenvalue  $\lambda$ , solves  $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$  for the corresponding eigenvector  $\mathbf{v}$ .
  - Any  $k$  eigenvectors  $\mathbf{v}^1, \dots, \mathbf{v}^k$  of  $\mathbf{A}$  with distinct eigenvalues  $\lambda^1, \dots, \lambda^k$  respectively, are linearly independent.
8. If  $\mathbf{A}$  has  $n$  linearly independent eigenvectors  $\mathbf{v}^1, \dots, \mathbf{v}^n$  with eigenvalues  $\lambda^1, \dots, \lambda^n$  (need not be distinct), then

$$\mathbf{x}^j(t) = e^{\lambda_j t} \mathbf{v}^j, \quad j = 1, \dots, n$$

are  $n$  linearly independent solutions of (4). And every solution  $\mathbf{x}(t)$  of (4) is of the form

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}^1 + c_2 e^{\lambda_2 t} \mathbf{v}^2 + \cdots + c_n e^{\lambda_n t} \mathbf{v}^n.$$

9. If  $\lambda = \alpha + i\beta$  is a complex eigenvalue of  $\mathbf{A}$  with complex eigenvector  $\mathbf{v} = \mathbf{v}^1 + i\mathbf{v}^2$ , then

$$\mathbf{x}(t) = e^{\alpha t} (\cos \beta t + i \sin \beta t) (\mathbf{v}^1 + i\mathbf{v}^2)$$

is a complex solution of (4). Furthermore,

$$\mathbf{y}(t) = \operatorname{Re}\{\mathbf{x}(t)\} = e^{\alpha t} (\mathbf{v}^1 \cos \beta t - \mathbf{v}^2 \sin \beta t)$$

$$\mathbf{z}(t) = \operatorname{Im}\{\mathbf{x}(t)\} = e^{\alpha t} (\mathbf{v}^1 \sin \beta t + \mathbf{v}^2 \cos \beta t)$$

are two independent real-valued solutions of (4).

10. A matrix  $\mathbf{X}(t) = [\mathbf{x}_1(t) \ \mathbf{x}_2(t) \ \cdots \ \mathbf{x}_n(t)]$  is called a *fundamental matrix solution* of (4) if its columns  $\mathbf{x}_1(t), \mathbf{x}_2(t), \dots, \mathbf{x}_n(t)$  form a set of  $n$  linearly independent solutions of (4).

11. The matrix-valued function  $e^{\mathbf{A}t}$  is defined as

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2 t^2}{2!} + \cdots + \frac{\mathbf{A}^n t^n}{n!} + \cdots$$

- $e^{\mathbf{A}t}$  is any fundamental matrix solution of (4).
- Let  $\mathbf{X}(t)$  be a fundamental matrix solution of (4), then

$$e^{\mathbf{A}t} = \mathbf{X}(t)[\mathbf{X}(0)]^{-1}.$$

12. Consider the system of nonhomogeneous linear differential equations with constant coefficients:

$$\dot{\mathbf{x}} = \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{f}(t), \quad \mathbf{x}(t_0) = \mathbf{x}^0. \quad (5)$$

Let  $\mathbf{X}(t)$  be any fundamental matrix solution of  $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ , then the solution of the initial-value problem (12) is given by

$$\mathbf{x}(t) = \mathbf{X}(t)[\mathbf{X}(t_0)]^{-1}\mathbf{x}^0 + \mathbf{X}(t) \int_{t_0}^t [\mathbf{X}(s)]^{-1}\mathbf{f}(s)ds.$$

In particular, take  $\mathbf{X}(t) = e^{\mathbf{A}t}$ , one has

$$\mathbf{x}(t) = e^{\mathbf{A}(t-t_0)}\mathbf{x}^0 + \int_{t_0}^t e^{\mathbf{A}(t-s)}\mathbf{f}(s)ds.$$